

Flutter OTP Resend Timer Documentation

Overview

This document explains an **OTP Resend Timer** implementation in Flutter. The timer prevents users from resending an OTP immediately and enables the **Resend code** action after 30 seconds.

This documentation is **fully editable, viewable**, and can be exported or copied anytime.

Features

- 🕒 Starts a 30-second countdown automatically
 - 🕒 Updates UI every second
 - 🔒 Disables resend action until timer ends
 - 🔧 Restarts timer when user taps **Resend code**
 - 🧹 Safely disposes the timer to avoid memory leaks
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State Variables Explained

Variable	Type	Purpose
<code>_timer</code>	<code>Timer?</code>	Runs every second to update countdown
<code>_secondsRemaining</code>	<code>int</code>	Remaining time before resend allowed
<code>_canResend</code>	<code>bool</code>	Controls resend button visibility

Full Source Code

```
async.Timer? _timer;  
int _secondsRemaining = 30;  
bool _canResend = false;  
  
@override  
void initState() {  
  super.initState();  
  startTimer();  
}  
  
void startTimer() {
```

```

_timer?.cancel();
_secondsRemaining = 30;
_canResend = false;

_timer = async.Timer.periodic(const Duration(seconds: 1), (timer) {
  if (_secondsRemaining == 0) {
    setState(() {
      _canResend = true;
    });
    timer.cancel();
  } else {
    setState(() {
      _secondsRemaining--;
    });
  }
});

}

@override
void dispose() {
  _timer?.cancel();
  otpController.dispose();
  super.dispose();
}

```

UI Usage Example

```

_canResend
? InkWell(
  onTap: () {
    startTimer();
  },
  child: Text(
    "Resend code",
    style: TextStyle.textStyle14w500c212B36sora
      .copyWith(color: AppColors.c3B82F6),
  ),
)
: RichText(
  text: TextSpan(
    text: "You can resend the code in ",
    style: TextStyle.textStyle14w500c919EABInter,
    children: [
      TextSpan(

```

```
        text: "$_secondsRemaining",
        style: TextStyle.textStyle14w500c212B36sora
          .copyWith(fontWeight: FontWeight.w700),
      ),
      TextSpan(
        text: " seconds",
        style: TextStyle.textStyle14w500c919EABInter,
      ),
    ],
  ),
);
```

How It Works (Step-by-Step)

1. Timer starts in `initState()`
2. Countdown decreases every second
3. UI updates using `setState()`
4. When time reaches zero, resend becomes active
5. Tapping **Resend code** restarts the timer

Notes & Best Practices

- Always cancel timers in `dispose()`
- Avoid multiple active timers by cancelling before restarting
- This logic is ideal for OTP, email verification, and SMS flows

 You can now **edit this document**, **copy code**, or **export it** anytime.